

Vibration-insensitive low coherence interferometer (LCI) for the measurement of technical surfaces



Yinan Li ^{*}, Markus Kästner, Eduard Reithmeier

Gottfried Wilhelm Leibniz Universität Hannover, Institute of Measurement and Automatic Control, Nienburger Str. 17, 30167 Hannover, Germany

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ABSTRACT

For the evaluation of operationally stressed surfaces containing microstructures, robot guided measurement systems need to be designed, that are able to automatically measure the integrity of these microstructures in a production related environment. Because drifts or vibrations between the robot mounted sensor and the sample surface noticeably influence measurement results, leading to usually insufficient results even large non-measured areas. The measurement system should be designed both to precisely measure the sample surface and to robustly handle the influences of the environmental vibration.

In this paper, we introduce a robot guided low coherence interferometer (LCI) system, that can compensate inevitable drifts and vibrations using high-speed data acquisition and a phase correction method, which can analyze and correct distorted interferograms using Hilbert-Transform in the frequency domain. Utilizing these methods the measurement performance of this system can be improved in comparison with the conventional LCI, to reduce non-measured areas and to reach a low measurement deviation of less than 0.1 μm . The detailed capability of the proposed methods for the identification and the compensation of environmental vibration is shown based on experimental data.

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1. Introduction

A technical surface containing microstructures is often crucial for maintaining the intended purpose. For example, micropores and grooves of cylinder liners serve as oil reservoirs to improve the tribological properties [1]. Riblets on compressor blades can reduce viscous drag in turbulent boundary layers [2]. While new parts are manufactured with a known machining process, the microgeometry is being measured often only for a small sample of the produced parts in a temperature controlled and vibration isolated measuring room. For maintenance and repair, each worn part has to be tested due to the only partly known history of wear effects. To limit the time consumption of the inspection the surface measurement technology has to be brought to the shop floor and ideally even be feasible for the inspection of still mounted parts.

Our approach for a fast accurate surface inspection focuses on the use of a well known measurement principle: low coherence interferometry (LCI). For the needed flexibility such devices are often mountable to an industrial six axes robot [3–5]. However, a immanent question is: How to perform an accurate measurement

using such a sensitive device in noisy environments? Some previous studies have been performed for similar applications. Seewig et al. studied a robust evaluation of intensity curves measured by confocal microscopies against vibration based on the Cramér-Rao bound [6]. Schmit et al. gave a real-time scanner error correction approach for high-precision shape measurement by white-light interferometry using a closed-loop system [7]. In the literature robot-based high-precision measurements using anti-vibration algorithms are rarely addressed.

In our case, the drifts or vibrations between a robot guided LCI and a sample surface are inevitable, because the mechanical connection from the sensor to the robot via the baseplate to the holding and lineally to the sample can be more than 2 m. In practical applications the environmental vibrations include different amplitudes and frequencies. To compensate these vibrations and to reduce the influences of these vibrations, hardware- and software-approaches are applied in this paper. They differ from the conventional anti-vibration approaches, which compensate vibrations mostly in a narrow frequency band and limited amplitudes using a single method. To deal with these issues our research concentrates on three approaches:

^{*} Corresponding author.

E-mail address: yinan.li@imr.uni-hannover.de (Y. Li).

- Design and construction of a LCI measurement system with high-speed data acquisition: utilizing a short scanning time the LCI is insensitive against low frequency drifts.
- Optimization of the setup of the robot mounted LCI system utilizing a optimized mechanical contact between LCI and robot, and avoiding the warm up phase of the robot: vibrations which are larger than the wavelength of the light source can be effectively damped.
- Analysis and correction of the obtained interferograms: the distorted interferograms whose amplitudes of the vibrations are less than the wavelength of the light source can be corrected using a Hilbert-transform based phase correction.

2. Experimental setup and principle

2.1. The low coherence interferometer

The inspection system presented in this paper is based on a low coherence Michelson interferometer (LCI) that is designed specifically for the purpose of vibration-insensitive surface measurement. The light source, a deep red light-emitting diode (LED), has a mean Gaussian spectrum with wavelength of 645 nm and a spectral bandwidth of 25 nm. The advantage of the red light is the narrow wave length spectrum which leads to a large coherence length, so that the constructive interferogram can be used a phase correction method, to minimize the influence of vibrations on the signal. More details about this method are given in Section 3.2.

Fig. 1 shows the experimental setup of the LCI system, which will be mounted on the robot arm. The beam from the LED light source is collimated using a Köhler illumination [8] to produce a homogeneous illumination, and is then splitted by a beam splitter into two beams that are perpendicular to each other with an equal intensity. The measuring beam is reflected by the sample surface and propagated through an objective to a camera. To ensure fast data acquisition, a Mikrottron MC-1362 high-speed camera and a frame grabber card are employed, which allows a resolution of 800×600 pixels with 8-bit gray-depth per 930 fps in Cameralink Full mode. The used compact telecentric objective has a very low image distortion degree and a $4\times$ magnification so that the LCI

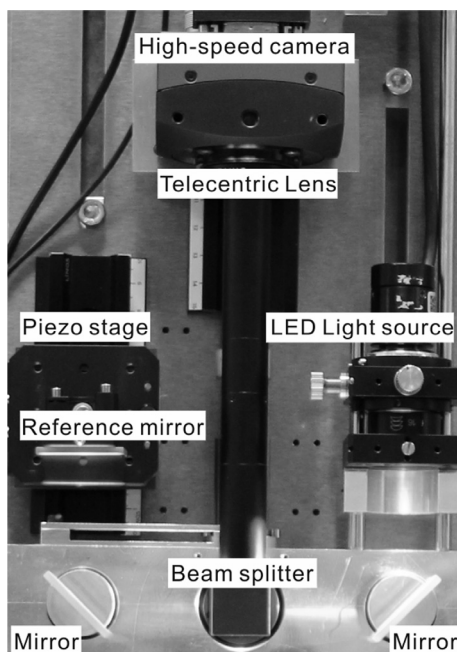


Fig. 1. The optical setup of the LCI.

can be insensitive to horizontal vibrations. The reference beam is reflected by a front face mirror, which is driven by a nano-positioning linear piezo stage. It controls the length of the reference arm of the interferometer in the measurement procedure. If the reference mirror and sample surface have the same distance from the beam splitter and they are also within the focus range of the objective, the two incident interfering beams are superposed and interfere with each other on the camera detector. The used high-speed monochrome camera which acquires up to 930 images per second can reduce the data acquisition time against low-frequency drifts [9]. Because of the very short exposure time periods, the LED illumination must work with high power. Nevertheless, for some very rough or non-flat surfaces the obtained images from LCI can be dark or very noisy. In this case, the frame rate of the camera should be reduced to increase the exposure time to guarantee clear images.

By changing the length of the reference arm using the piezo stage in a very small range, a data stack of the obtained intensity images is recorded simultaneously by the camera for calculating the height of the sample surface. Each of the pixels in the data stack in z direction corresponds to a scanned signal that contains information about intensity variations with changed optical path difference (OPD) between reference and measurement arms (see Fig. 2). The maximum intensity of the envelope of the signal corresponds to the length difference of the reference arm (also relative height). As signal processing, the envelope of the signal is calculated using the Hilbert-Transform (HT) and then the peak positions of the envelopes are recorded as the relative surface height [10]. The lateral resolution is determined by the combination of the pixel size, the optical aperture of the camera and objective magnification chosen for different fields of view (FOV). A lateral resolution of $7 \mu\text{m}$ results in a field of view of about $1.7 \times 2.2 \text{ mm}^2$.

2.2. Setup of the robot guided LCI system

In order to guarantee a high flexibility and accessibility for the automated inspection of surface microstructures, the measurement system permits all six degrees of freedom (DOF) of the movement. The presented system shown in Fig. 3 consists of an industrial six axes robot arm and an in-house developed LCI. The LCI is hanged on the end joint of the robot and the samples are placed on an additional motorized rotation stage that is damped by a vibration isolating table. The additional rotational axis reduces the necessary movement of the positioning system, so that vibration influences by transition of inspection positions can be

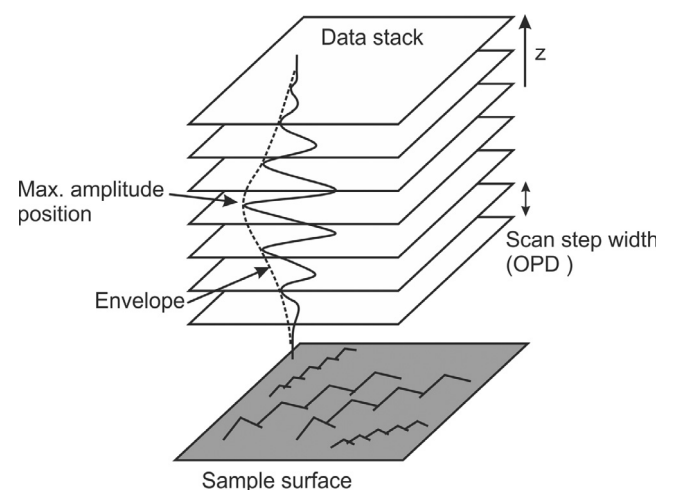


Fig. 2. The data structure of the obtained interference patterns.

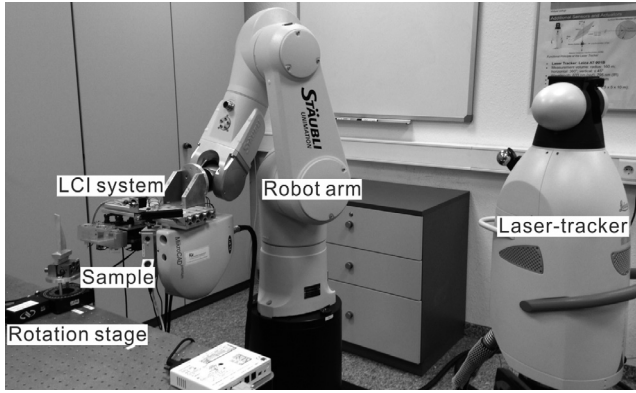


Fig. 3. The robot guided LCI measurement system.

decreased. The main drawback of the robot guided LCI system is that it is easily affected by environment noise and disturbances like thermal drift and mechanical vibrations. To isolate the multi-directional vibrations, the mechanical contact between the robot and the LCI system is replaced by a soft rubber damping set made by DJI Innovations which can isolate most vertical and horizontal vibrations. This construct can be seen as a spring-damping system with a large damping factor, which can effectively isolate the large-amplitude vibrations.

2.3. Estimation and reduction of the expected drift

Because it is nearly impossible to estimate the drift of an industrial robot in a scale below 100 nm a priori, a Leica AT-901 laser-tracker shown in Fig. 3 is used to measure the relative drift between the sensor head of the robot guided LCI and the sample in axial direction. The position and orientation of the end-effectors of the robot was measured every 200 ms over several minutes and directly after turning on the motors to get an impression of the expected scale of drifts.

The maximum drift of the robot measured by the laser-tracker was approx. 97 nm/s during turning on and approx. 36 nm/s over 15 min holding the static pose shown in Fig. 4. During the first 15 min the robot arm in the warm up phase drifts with larger amplitude than in the steady state. To consider the effect of low frequency drifts after the warm up phase, the scanning time of the LCI should be less than 3 s to reduce the effect of the drift. In this case, the LCI is adjustable by more than 900 fps sampling frequency.

3. Estimation and reduction of the effect of vibrations

In the previous section, some anti-vibration hardware-based methods like high-speed data acquisition and mechanical damping

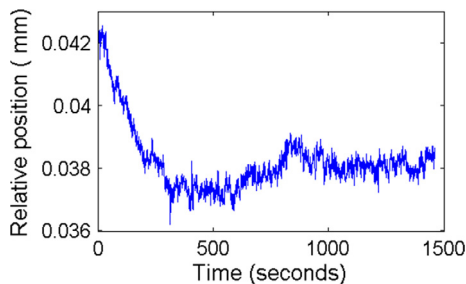


Fig. 4. The measured drift between LCI sensor and sample surface using a Leica Laser-tracker.

are presented. To a large extent, low-frequency distortions like drifts can be reduced utilizing these methods. The further software-based phase correction method to lessen vibrations is described in this section.

3.1. Simulation of interferograms using the spectrum of the applied light source

The computer simulation of interferograms provides a mathematical model of a quasi-monochromatic LED light source for one pixel of the CCD camera. The basic intensity equation of the interferogram for the LED light source is defined:

$$I(z) = I_0 + I_1 \cos(kz + \varphi_0) e^{-((z-z_0)^2/2w_g^2)}, \quad (1)$$

where $I(z)$ is the intensity value at the position z , φ_0 is the inner phase shift, $e^{-((z-z_0)^2/2w_g^2)}$ is the Gaussian envelope of the interference, z_0 the peak position of the envelope and w_g the width of the envelope. In the scanning procedure, the length of the measurement arm or the OPD value z is consistently changed with a scan step Δz , which is driven by the piezo stage. The correlogram is sampled at the scan positions $z = n\Delta z$, where n is the frame number of the camera. Moreover, in order to simulate vibrations in the direction of the measurement arm, a random phase shift $z_{ran}(n)$ that occurs at the same time of the frame number n is added into the equation of the interference. The phase error is accumulated as the sum of $\sum_0^n z_{ran}(n)$ into the correlogram, whose expression is

$$I(n\Delta z) = I_0 + I_1 \cos \left(k \left(n\Delta z + \sum_0^n z_{ran}(n) \right) + \varphi_0 \right) e^{-((n\Delta z + \sum_0^n z_{ran}(n) - z_0)^2/2w_g^2)}. \quad (2)$$

Fig. 5 shows the simulated correlograms, one in Fig. 5(a) without phase shifts and the other one in Fig. 5(b) with random phase shifts, where the parameters are set to $\varphi_0 = 0$ and $n = 1000$.

3.2. Signal analysis using the discrete Hilbert transform

To correct the simulated correlogram with undesirable phase shifts, a phase correction algorithm based on the discrete Hilbert transform (DHT) is used [11]. For the discrete-time sampled correlogram, the DHT can generate Fourier transform components, and in consequence:

$$H(u[n]) = DTFT^{-1}(U(w) \cdot \sigma_H(w)), \quad (3)$$

where $u[n]$ is the sampled correlogram with the frame number n , $U(w)$ is the value of the discrete-time Fourier transform (DTFT) and $\sigma_H(w)$ is a signum function in the form:

$$\sigma_H(w) = \begin{cases} e^{+i\pi/2} & -\pi < w < 0 \\ e^{-i\pi/2} & 0 < w < \pi \\ 0 & w = -\pi, 0, \pi \end{cases}. \quad (4)$$

Utilizing the inverse DTFT of $U(w) \cdot \sigma_H(w)$ the analytic representation of the Hilbert signal can be constructed in this form:

$$\hat{U}[n] = u[n] + i \cdot H(u[n]), \quad (5)$$

where the real part is the correlogram and the imaginary part is its Hilbert transform with $\pi/2$ phase shift. The Hilbert transform allows extracting the inner phase of the correlogram and the envelope of the whole correlogram [12]. The phase and the envelope of the signal can be defined by the equations:

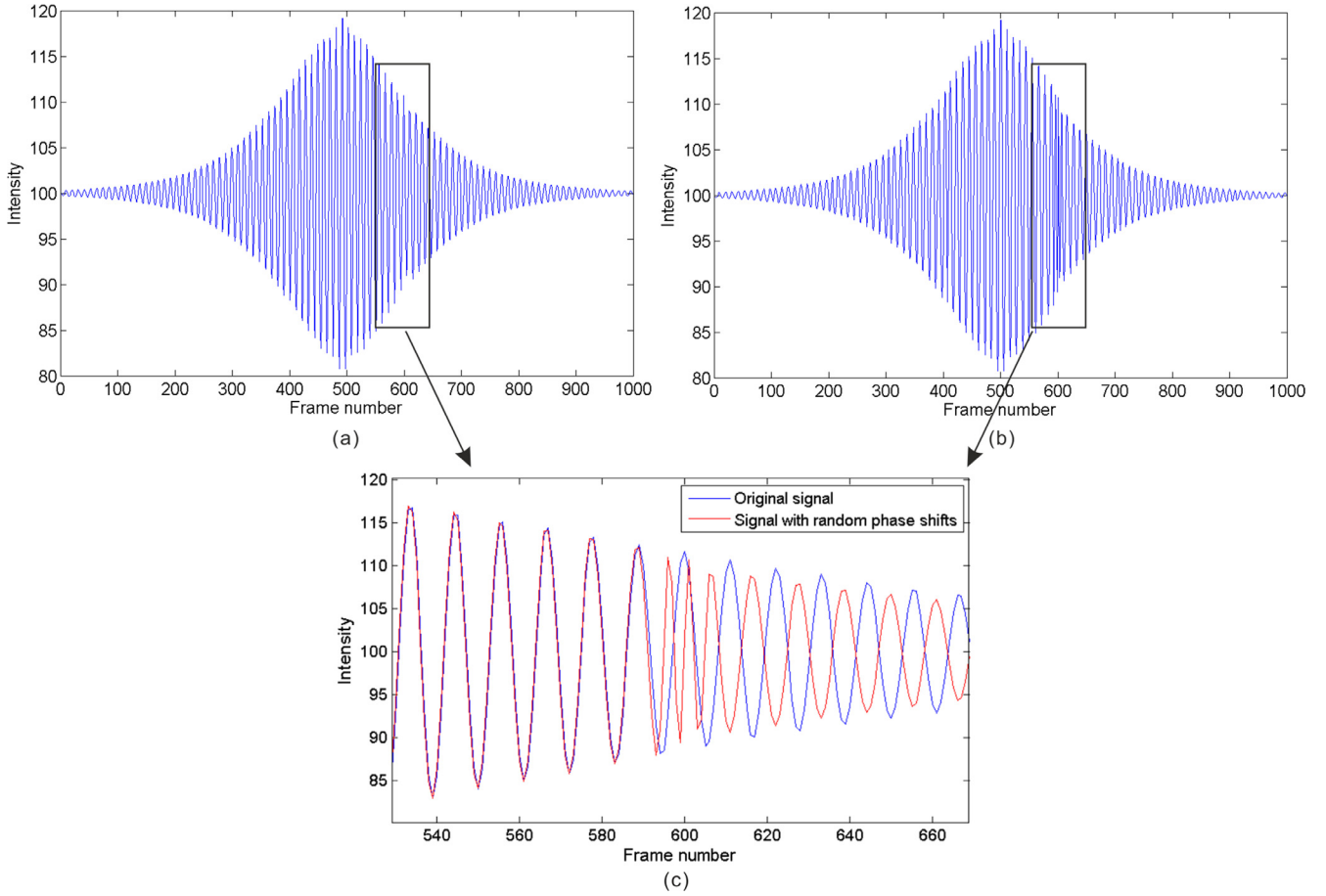


Fig. 5. The simulations of the correlogram: (a) original signal, (b) signal with random phase shifts, (c) comparison of signals.

$$\psi(n\Delta z) = \arctan \frac{H(u[n])}{u[n]} \quad \text{and} \quad A(n\Delta z) = \sqrt{u[n]^2 + H(u[n])^2}. \quad (6)$$

For calculating the phase shift and the envelope of the correlogram, the fundamental intensity I_0 in the correlogram equation should be removed. Hence, the phase ψ can be found using Eq. (2) without the offset I_0 like:

$$\begin{aligned} \psi(n\Delta z) &= \arctan \frac{H(u[n])}{u[n]} \\ &= \arctan \frac{I_1 \sin \left(k \left(n\Delta z + \sum_0^n z_{ran}(n) \right) + \varphi_0 \right) e^{-((n\Delta z + \sum_0^n z_{ran}(n) - z_0)^2 / 2w_g^2)}}{I_1 \cos \left(k \left(n\Delta z + \sum_0^n z_{ran}(n) \right) + \varphi_0 \right) e^{-((n\Delta z + \sum_0^n z_{ran}(n) - z_0)^2 / 2w_g^2)}} \\ &= k(n\Delta z + \sum_0^n z_{ran}(n)) + \varphi_0. \end{aligned} \quad (7)$$

To further simplify the relationship of Eq. (7), we assume that the piezo stage in the LCI moves stably in z direction and its scan step Δz in the measurement procedure is unchanged. For the compensation of the phase shift error the frame number n is defined in this paper as a variable, therefore the corrected frame number for each sampling point $\hat{n}$ can be described as:

$$\hat{n} = \frac{\psi(n\Delta z) - \varphi_0}{k\Delta z} - \frac{\sum_0^n z_{ran}(n)}{\Delta z}. \quad (8)$$

In this way, if the random phase shift $z_{ran}(n)$ remains unchanged in the scanning procedure, it means that no vibrations occur in z direction, thus the calculated frame number n is a constant. Otherwise, if the variable $z_{ran}(n)$ changes, the phase change of n is non-linearly monotonically increasing. It means that the real sampling step size is not a constant due to vibrations in z direction. Fig. 6(a) compares the variations of frame number between original signal without phase shift error and the signal with phase shift error. Moreover, Fig. 6(b) shows the comparison of the signals with and without the phase correction. The phase difference of the correlograms before and after the phase correction is very small. Moreover, the peak position z_p and the corresponding frame numbers n_p can be estimated from:

$$\begin{aligned} z_p &= \max_z (A(\hat{n}\Delta z)) = \max_z (I_1 e^{-((\hat{n}\Delta z + \sum_0^{\hat{n}} z_{ran}(\hat{n}) - z_0)^2 / 2w_g^2)}) = z_0 - \sum_0^{\hat{n}} z_{ran}(\hat{n}), \\ n_p &= \frac{z_0 - \sum_0^{\hat{n}} z_{ran}(\hat{n})}{\Delta z}. \end{aligned} \quad (9)$$

If the measurement procedure is undisturbed, therefore the random phase shift is $z_{ran}(n) = 0$, the detected peak position z_p equals the ideal position z_0 . However, the obtained peak position $\hat{z}$ contains accumulated phase shift errors $\sum_0^{\hat{n}} z_{ran}(\hat{n})$. In this case, the phase shift correction is necessary for extracting the exact peak position $\hat{z}$ of a correlogram.

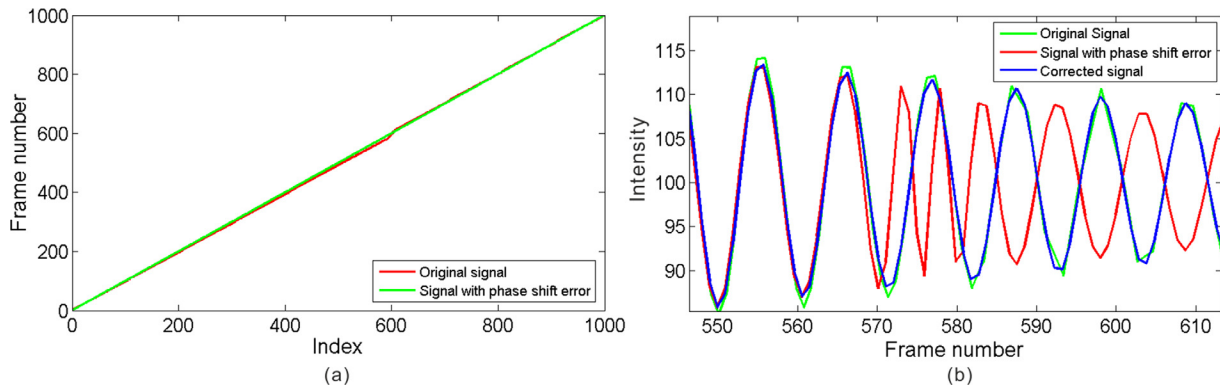


Fig. 6. The phase analysis of the correlogram: (a) comparison of the phase between original signal and signal with phase shift error, (b) comparison of the signals before and after the phase correction.

4. Results

In order to analyze the performance of the robot guided LCI measurement system to compensate occurring vibrations and drifts, a flatness standard as a low roughness reference surface was measured at first. The obtained intensity image stacks were evaluated by the, in Section 3.2, proposed method for phase correction and estimation of the surface height. The measurements were carried out at two different sampling frequencies, firstly with a sampling frequency of 50 fps to simulate a conventional interferometer, and then with a sampling frequency of 930 fps. Afterwards the recorded data was evaluated with and without phase correction respectively.

Fig. 7 shows the measured disturbed surface topographies of the flatness standard. The measurement, which was recorded with the sampling frequency of 50 fps requires more scanning time than high speed sampling (930 fps) and leads to more signal distortions in the sampled data due to drifts and vibrations, so that several areas of the sample surface were not correctly evaluated, not even

measured. Fig. 7(a) shows strong distortions with a standard deviation of $2.2 \mu\text{m}$. Correspondingly, the evaluated surface, that was acquired with the high speed sampling seen in Fig. 7(b), fluctuates in a relatively lower undulation with a standard deviation of $0.96 \mu\text{m}$. The direction of this undulation is parallel to the direction of the interference patterns in the image stacks. This means that the observed undulation is caused by vibrations during the measurement procedure. In this case, the evaluation method with phase correction offers an effective improvement to lessen the effects of vibrations on the measurement results (see Fig. 7(c)). The standard deviation of the surface with phase correction is less than $0.35 \mu\text{m}$.

Although the effects of vibrations remain in this results, the LCI system allows an improved accuracy for the practical applications. Fig. 8 shows the results of measurements of a roughness standard with a roughness average $R_a = 6 \mu\text{m}$ under three different measurement conditions. Due to an increasing vertical scanning range and the corresponding longer measurement time, the scanned surface with the 50 fps sampling frequency contains more non-

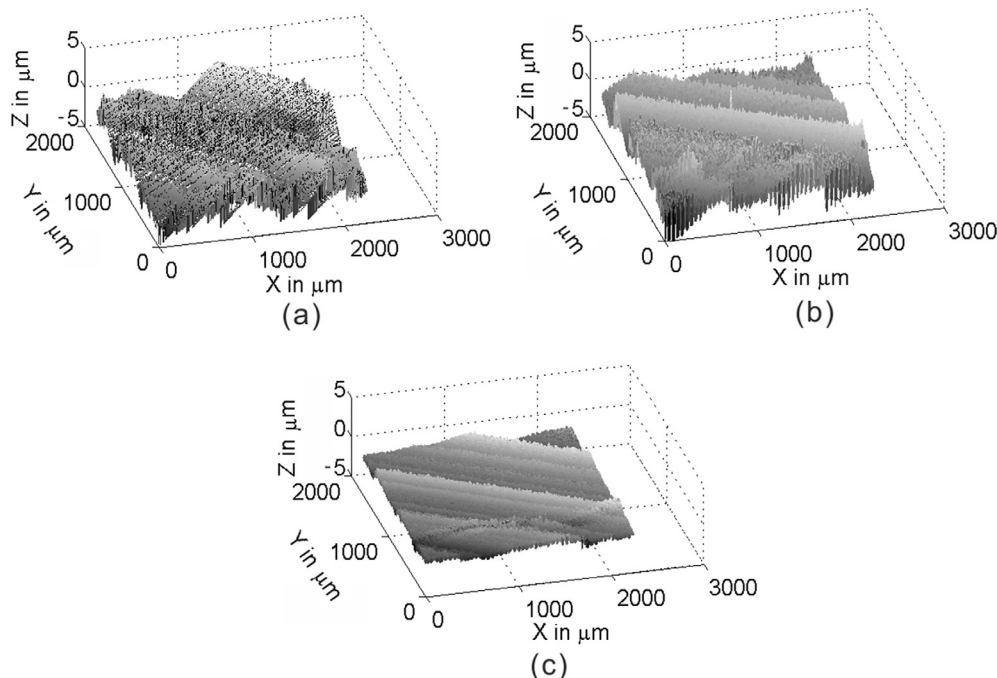


Fig. 7. Measured flatness standard in the disturbed measurement produce: (a) data evaluation for low frequency sampling measurement, (b) data evaluation without phase correction, (c) data evaluation with phase correction.

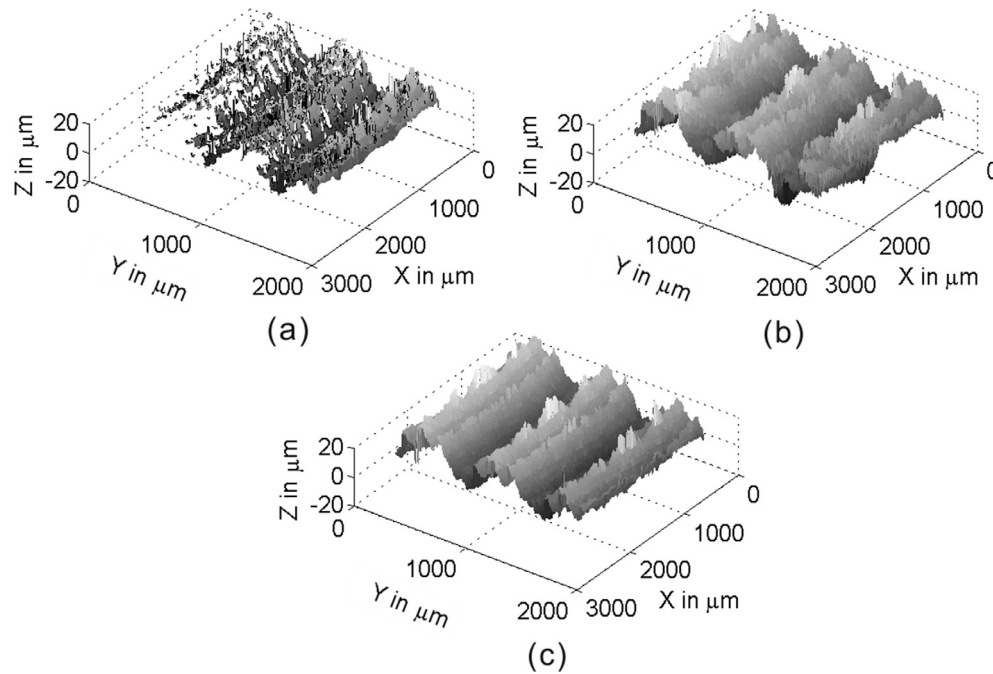


Fig. 8. Measurement of roughness standard: (a) data evaluation for low frequency sampling measurement, (b) data evaluation without phase correction, (c) data evaluation with phase correction.

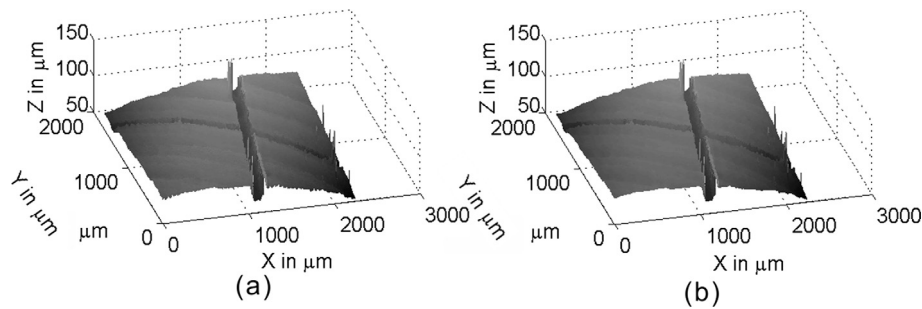


Fig. 9. Measurement of damage on a turbine blade: (a) data evaluation without phase correction, (b) data evaluation with phase correction.

evaluated areas (see Fig. 8(a)). The scanned surfaces with high speed sampling contain very low distortions (see Fig. 8(b and c)). The calculated R_a of the evaluated surface with low sampling frequency is greater than $15\ \mu\text{m}$. The R_a of the evaluated surface without phase correction is $7.15\ \mu\text{m}$ and the R_a with phase correction is even more accurate $6.45\ \mu\text{m}$.

Further, to test the applicability of the measurement system for a large scanning range, a surface defect on a turbine blade was measured. Because the scanned surface has high curvatures, the corresponding vertical scanning range was set to $100\ \mu\text{m}$. In this situation, the low sampling frequency can lead to a very large non-measured area. Therefore, we compared only the measurement results with high speed sampling, seen in Fig. 9. The difference between the data evaluation with and without phase correction is not obvious due to the large scaling. In this situation both methods offer almost identical results for the measured width and depth of the selected cracks on the surface.

5. Conclusion and outlook

This paper introduces a robot guided LCI measurement system, which allows the measurement of surface topographies and the compensation of occurring vibrations and drifts. The advantage of

this system is that the high-speed image data acquisition can reduce the scanning time with regard to the low frequency drift and further the phase correction method can correct disturbed correlograms with regard to higher frequency vibrations. Utilizing these advantages, it is possible to reach a low measurement deviation of less than $0.1\ \mu\text{m}$ by using the proposed evaluation method in comparison to a conventional measurements. Moreover, for a large vertical scanning range, the LCI system leads to a more robust evaluation of the surface topography by reducing the non-measured area.

The vibrations and drifts in the current measurement environment are less than $36\ \text{nm/s}$. If under poor measurement conditions the distortions become stronger with an increasing amplitude or frequency, the performance of the phase correction gets worse. In this situation, the measurement accuracy of the LCI system can be improved using an active scanner correction with a closed-loop vibration compensation [13,14].

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